

REACT AND SPRING-BOOT&MYSQL

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REACT AND SPRING-BOOT&MYSQL

1. Api

insert

Post:- <http://localhost:8084/api/v1/addStudent>

Body->raw -> json:-

```
{
  "name": "rk",
  "age": 21,
  "dept": "Computer Science"
}
```

List

Get:- <http://localhost:8084/api/v1/getStudents>

getById:-

get:- <http://localhost:8084/api/v1/studentById/10>

delete:-

delete :- <http://localhost:8084/api/v1/deleteStudent/2>

update

put :- <http://localhost:8084/api/v1/updateStudent/3>

body->raw -> Json ->

```
{  
  
  "name": "abhi ",  
  "age": 20,  
  "dept": "Computer Science"  
}
```

2. BACKEND

Link is here a:-

<https://github.com/Ranjitk0001/React/tree/react/Projects/ReactAndSpringBootCrudOpr/reactspringbootBackend>

Spring start io (create project and dependency :- Lombok,spring web,spring web tool,mysql ,jpa)

Entity, Repository, Services , Controller there are Four package main

Entity package in entity.java class

```
package com.reactspringboot.reactspringboot.entity;
```

```
import jakarta.persistence.*;  
import lombok.Data;
```

```
//@Data
```

```
@Entity
```

```
  @Table(name ="student")
```

```
  public class Student {
```

```
    @Id
```

```
    @GeneratedValue(strategy = GenerationType.IDENTITY)
```

```

    private int id;
    @Column(name="student_name")
    private String name;
    private int age;
    @Column(name="department")
    private String dept;

    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }
}

```

using getter ,setter, Constructor, or many..

Repository package in StudentRepository.java class

```
package com.reactspringboot.reactspringboot.repository;
```

```

import com.reactspringboot.reactspringboot.entity.Student;
import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.stereotype.Repository;

```

```

@Repository
public interface StudentRepository extends JpaRepository<Student,Long> {

}

```

Services package in StudentService.java class

```
package com.reactspringboot.reactspringboot.services;
```

```

import com.reactspringboot.reactspringboot.entity.Student;
import com.reactspringboot.reactspringboot.repository.StudentRepository;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

```

```

import javax.management.AttributeNotFoundException;
import java.util.List;

```

```

@Service
public class StudentService {
    @Autowired

```

```

private StudentRepository studentRepository;

public Student addStudent(Student student) {
    return studentRepository.save(student);
}

public List<Student> getAllStudents() {
    return studentRepository.findAll();
}

public Student getStudentById(long id) throws AttributeNotFoundException {
    return studentRepository.findById(id)
        .orElseThrow(() -> new AttributeNotFoundException("Student not found with id " + id));
}

public Student updateStudent(long id, Student updatedStudent) throws AttributeNotFoundException {
    Student student = studentRepository.findById(id) .orElseThrow(() -> new
AttributeNotFoundException("Student not found with id " + id));
    student.setAge(updatedStudent.getAge());
    student.setDept(updatedStudent.getDept());
    student.setName(updatedStudent.getName());
    return studentRepository.save(student);
}

public void deleteStudent(long id) throws AttributeNotFoundException {
    Student student = studentRepository.findById(id) .orElseThrow(() -> new
AttributeNotFoundException("Student not found with id " + id));
    studentRepository.delete(student);
}
}

```

Controller package in StudentController.java class

```

package com.reactspringboot.reactspringboot.controller;

import com.reactspringboot.reactspringboot.entity.Student;
import com.reactspringboot.reactspringboot.services.StudentService;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.*;

import javax.management.AttributeNotFoundException;
import java.util.HashMap;
import java.util.List;
import java.util.Map;

@RestController
@CrossOrigin("*")
@RequestMapping("/api/v1")
public class StudentController {
    @Autowired
    private StudentService studentService;

```

```

@PostMapping("/addStudent")
public Student addStudent(@RequestBody Student student) {
    if (student == null)
        { throw new RuntimeException("StudentService is not injected properly");}
//    System.out.println("Student Data :- ====="+student.toString());
    System.out.println("Received student data: " + student);
    Student savedStudent = studentService.addStudent(student);
    return studentService.addStudent(student);
}

@GetMapping("/getStudents")
public List<Student> getAllStudents(){
    return studentService.getAllStudents();
}

@GetMapping("/studentById/{id}")
public ResponseEntity<Student> studentById(@PathVariable long id) throws AttributeNotFoundException
{
    Student student = studentService.getStudentById(id);
    return ResponseEntity.ok(student);
}

@PutMapping("/updateStudent/{id}")
public ResponseEntity<Student> updateStudentC(@PathVariable long id, @RequestBody Student
updatedStudent) throws AttributeNotFoundException {
    Student savedStudent = studentService.updateStudent(id, updatedStudent);
    return ResponseEntity.ok(savedStudent);
}

@DeleteMapping("/deleteStudent/{id}")
public ResponseEntity<Map<String, Boolean>> deleteStudent(@PathVariable int id) throws
AttributeNotFoundException {
    studentService.deleteStudent(id);
    Map<String, Boolean> response = new HashMap<>();
    response.put("Deleted", Boolean.TRUE);
    return ResponseEntity.ok(response);
}
}

```

Application.properties all configuration in database and hibernate
spring.application.name=reactspringboot
server.port=8084

DataSource Configuration

spring.datasource.url=jdbc:mysql://localhost:3306/ReactSpringBootCrudOpr?createDatabaseIfNotExist=true

spring.datasource.username=root

spring.datasource.password=Pass@123

JPA Configuration

spring.jpa.hibernate.ddl-auto=update

spring.jpa.show-sql=true

spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.MySQL8Dialect

3. FRONTEND

Link is here: _

<https://github.com/Ranjitk0001/React/tree/react/Projects/ReactAndSpringBootCrudOpr/react-spring-boot-frontent>

REACT JS AND BOOTSTRAP

Create react app and install bootstrap and axios for apis fetching and router-dom install

Index.js add bootstrap link here

Services it is using for bussness logic in some opearation (api fetch & all apis is here)

import axios from 'axios'

const BASE_REST_API_URL = 'http://localhost:8084/api/v1';

class StudentService {

getAllStudents(){

return axios.get(BASE_REST_API_URL+'/getStudents');

}

createStudent(student){

return axios.post(BASE_REST_API_URL+'/addStudent', student);

}

deleteStudent(studentId){

return axios.delete(BASE_REST_API_URL + '/deleteStudent/' + studentId);

}

```

updateStudent(studentId, student){
  return axios.put(BASE_REST_API_URL + '/updateStudent/' +studentId, student);
}

getStudentById(studentId){
  return axios.get(BASE_REST_API_URL + '/studentById/' + studentId);
}

}

const instance = new StudentService();
export default instance;

```

Component it is used all pages here

RouteNavbar.js

```

import React from 'react'
import { NavLink } from 'react-router-dom'
const RouteNavbar = () => {
  return (
    <>
      <nav className="navbar navbar-expand-lg bg-body-tertiary">
        <div className="container-fluid">
          Navbar
          <button className="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target="#navbarSupportedContent" aria-controls="navbarSupportedContent" aria-expanded="false" aria-
label="Toggle navigation">
            <span className="navbar-toggler-icon"></span>
          </button>
          <div className="collapse navbar-collapse" id="navbarSupportedContent">
            <ul className="navbar-nav me-auto mb-2 mb-lg-0">
              <li className="nav-item">

```

```
<NavLink className="nav-link active" to={'/'}>Home</NavLink>
```

```
</li>
```

```
<li className="nav-item">
```

```
<NavLink className="nav-link active" to={'/add-student'}>Add Student</NavLink>
```

```
</li>
```

```
<li className="nav-item">
```

```
<NavLink className="nav-link active" >Disabled</NavLink>
```

```
</li>
```

```
</ul>
```

```
</div>
```

```
</div>
```

```
</nav>
```

```
</>
```

```
)
```

```
}
```

```
export default RouteNavbar
```

App.js

```
import './App.css';
```

```
import { BrowserRouter as Router, Routes, Route } from "react-router-dom";
```

```
import StudentList from './components/StudentList';
```

```
import StudentAdd from './components/StudentAdd';
```

```
import StudentUpdate from './components/StudentUpdate';
```



```

function App() {
  return (
    <>

      { /* <h1>Welcome to React Spring Boot</h1> */ }

      <Router>
      <Routes>
        <Route path="/" element={<StudentList />} />
        <Route path="/add-student" element={<StudentAdd />} />
        <Route path="/edit-student/:id" element={<StudentUpdate />} />
      </Routes>
    </Router>
  </>
);
}

```

```

export default App;

```

StudentList.js

```

import React, { useEffect, useState } from "react";
import StudentService from "../services/StudentService";
import RouteNavbar from "../RouteNavbar";
import { Link } from "react-router-dom";

const StudentList = () => {

```

```
const [students, setStudent] = useState([]);
```

```
useEffect(() => {  
  getAllStudents();  
}, []);
```

```
const getAllStudents = () => {  
  StudentService.getAllStudents()  
    .then((response) => {  
      setStudent(response.data);  
      console.log(response.data);  
    }).catch((error) => {  
      console.log(error);  
    });  
};
```

```
const deleteStudent = (studentId) => {  
  StudentService.deleteStudent(studentId).then((response) => {  
    getAllStudents();  
  }).catch((error) => {  
    console.log(error);  
  });  
};
```

```
return (  
  <>  
    <RouteNavbar />  
    <h1>Student List</h1>  
    <div className="container mt-5">  
      <h1>Bootstrap Table Example</h1>  
      <table className="table table-striped table-hover">  
        <thead>
```

```

    <tr>
      <th scope="col">Id</th>
      <th scope="col">Name</th>
      <th scope="col">Age</th>
      <th scope="col">Department</th>
      <th> Actions </th>
    </tr>
  </thead>
  <tbody>
    {
      students.map((student) => (
        <tr key={student.id}>
          <th scope="row">{student.id}</th>
          <td>{student.name}</td>
          <td>{student.age}</td>
          <td>{student.dept}</td>
          <td>
            <Link className="btn btn-info" to={` /edit-student/${student.id}`}>Update</Link>
            <button className="btn btn-danger" onClick={() => deleteStudent(student.id)} style={{
marginLeft: "10px" }}> Delete</button>
          </td>
        </tr>
      ))
    }
  </tbody>
</table>
</div>
</>

);
};

```

```
export default StudentList;
```

StudentAdd.js

```
import React, { useState } from 'react';
import { Link, useNavigate } from 'react-router-dom'; // Use useNavigate instead of useHistory
import RouteNavbar from './RouteNavbar';
import StudentService from '../services/StudentService';

const StudentAdd = () => {
  const [name, setName] = useState("");
  const [age, setAge] = useState("");
  const [dept, setDept] = useState("");
  const navigate = useNavigate(); // Use useNavigate

  const saveStudent = (e) => {
    e.preventDefault();
    const student = { name, age, dept };

    StudentService.createStudent(student).then((response) => {
      console.log(response.data);
      navigate('/');
    });
  };

  return (
    <>
      <RouteNavbar />
      <div className="container mt-5">
        <h1>Add Student</h1>
        <div className="row">
          <div className="card col-md-6 offset-md-3">
```

```
<div className="card-body">
  <form>
    <div className="form-group mb-2">
      <label className="form-label">Name:</label>
      <input
        type="text"
        placeholder="Enter name"
        name="name"
        className="form-control"
        value={name}
        onChange={(e) => setName(e.target.value)}
      />
    </div>
    <div className="form-group mb-2">
      <label className="form-label">Age:</label>
      <input
        type="text"
        placeholder="Enter age"
        name="age"
        className="form-control"
        value={age}
        onChange={(e) => setAge(e.target.value)}
      />
    </div>
    <div className="form-group mb-2">
      <label className="form-label">Department:</label>
      <input
        type="text"
        placeholder="Enter department"
        name="dept"
        className="form-control"
        value={dept}
      />
    </div>
  </form>
</div>
```

```

        onChange={(e) => setDept(e.target.value)}
      />
    </div>
    <button className="btn btn-success" onClick={(e) => saveStudent(e)}>Submit</button>
    <Link to="/" className="btn btn-danger">Cancel</Link>
  </form>
</div>
</div>
</div>
</div>
</>
);
}

```

```
export default StudentAdd;
```

StudentUpdate.js

```

import React, { useEffect, useState } from "react";
import { Link, useNavigate, useParams } from "react-router-dom";
import StudentService from "../services/StudentService";
import RouteNavbar from "../RouteNavbar";

const StudentUpdate = () => {
  const [name, setName] = useState("");
  const [age, setAge] = useState("");
  const [dept, setDept] = useState("");
  const navigate = useNavigate();
  const { id } = useParams();
  const saveStudent = (e) => {
    e.preventDefault();

```

```
const student = { name, age, dept };
```

```
if (id) {
```

```
  StudentService.updateStudent(id, student)
```

```
    .then((response) => {
```

```
      navigate("/");
```

```
    })
```

```
    .catch((error) => {
```

```
      console.log(error);
```

```
    });
```

```
}
```

```
};
```

```
useEffect(() => {
```

```
  StudentService.getStudentById(id)
```

```
    .then((response) => {
```

```
      setName(response.data.name);
```

```
      setAge(response.data.age);
```

```
      setDept(response.data.dept);
```

```
      console.log(response.data);
```

```
    })
```

```
    .catch((error) => {
```

```
      console.log(error);
```

```
    });
```

```
}, [id]);
```

```
return (
```

```
<>
```

```
<RouteNavbar />
```

```
<div className="container mt-5">
```

```
<div className="row">
```

```
<div className="card col-md-6 offset-md-3">
```

```
<div className="card-body">
```

```
<h1>Update Student</h1>

<form>

  <div className="form-group mb-2">

    <label className="form-label">Name:</label>

    <input

      type="text"

      placeholder="Enter name"

      name="name"

      className="form-control"

      value={name}

      onChange={(e) => setName(e.target.value)}

    />

  </div>

  <div className="form-group mb-2">

    <label className="form-label">Age:</label>

    <input

      type="text"

      placeholder="Enter age"

      name="age"

      className="form-control"

      value={age}

      onChange={(e) => setAge(e.target.value)}

    />

  </div>

  <div className="form-group mb-2">

    <label className="form-label">Department:</label>

    <input

      type="text"

      placeholder="Enter department"

      name="dept"

      className="form-control"

      value={dept}
```


